NTNU Department of Information Security and Communication Technology

TTM4135 Applied Cryptography and Network Security
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Worksheet 6: Discrete log algorithms and digital signatures

QUESTION 1

Review the definitions of the following concepts. They are things that you would be expected to know in the exam.

(a) discrete logarithm problem
(b) generator of $\mathbb{Z}_p^*$
(c) Diffie–Hellman key exchange
(d) Elgamal cryptosystem
(e) digital signature
(f) existential forgery and selective forgery
(g) digital signature algorithm (DSA)
(h) ECDSA

QUESTION 2

Let $p = 43$. Verify that $g = 3$ is a generator of $\mathbb{Z}_p^*$. Suppose that Alice and Bob execute the Diffie–Hellman key exchange protocol with Alice's input being $a = 5$ and Bob's input $b = 13$. Show that both Alice and Bob will compute the same shared secret.

QUESTION 3

The Diffie–Hellman protocol can be defined, but is not necessarily secure, in any group.

(a) Define Diffie–Hellman in the *additive* group modulo p for some prime p, instead of the multiplicative group where it is usually defined. Would this be secure for sufficiently large values of p?

(b) Write down the equations for Diffie-Hellman on elliptic curves (ECDH) using the notation on Slide 17 of Lecture 11. Show that for this to be secure the elliptic curve discrete log problem must not be an easy problem

QUESTION 4

It is common in the Elgamal encryption algorithm for users to share the modulus p and generator g. Why is it not possible for users to share the same modulus n in the RSA cryptosystem?

QUESTION 5

In the Elgamal cryptosystem Alice and Bob have public keys g^{x_A} and g^{x_B} respectively, with corresponding private keys x_A and x_B. When Alice wants to send a message confidentially to Bob she chooses an ephemeral private key a and constructs a new shared secret g^{ax_B}.

Consider the following variant of the Elgamal cryptosystem. Instead of choosing a new random value a, Alice simply computes the static Diffie–Hellman value $X = (y_B)^{x_A} \bmod p$ and sends $C = MX \bmod p$ to Bob as the ciphertext.

(a) How does Bob decrypt?
(b) What could be the advantages of such a scheme as compared with normal Elgamal encryption?
(c) Show that this scheme is, unfortunately, completely insecure against a known plaintext attack.

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QUESTION 6

Suppose that Alice has a public key $y = 5, g = 2, p = 11$ for the Elgamal encryption algorithm. Here $g = 2$ is a generator for $\mathbb{Z}_{11}^*$. Compute a valid ciphertext for the message $M = 3$ intended for Alice, showing the steps required.

QUESTION 7

Compare the efficiency of the Elgamal cryptosystem in $\mathbb{Z}_p^*$ and the RSA cryptosystem with modulus n. Assume that the size of the modulus p and n is the same in each case. Compare:

- the cost of key generation;
- the computation required for encryption;
- the computation required for decryption;
- the size of the public keys and ciphertexts.

QUESTION 8

In 2019 elections for the Moscow city parliament an electronic voting system used a simple variant of ElGamal encryption to protect a user vote M. For the public key $K = (K_1, K_2, K_3) = (y_1, y_2, y_3) = (g^{x_1}, g^{x_2}, g^{x_3})$ the following details the encryption algorithm. First the encrypting party chooses random k_1, k_2, k_3 and computes the following values.

$C_1 = E(M, K_1) = (g^{k_1} \bmod p, My_1^{k_1} \bmod p) = (a_1, b_1)$

$C_2 = E(a_1, K_2) = (g^{k_2} \bmod p, a_1 y_2^{k_2} \bmod p) = (a_2, b_2)$

$C_3 = E(a_2, K_3) = (g^{k_3} \bmod p, a_2 y_3^{k_3} \bmod p) = (a_3, b_3)$

Finally the ciphertext is the 4-tuple $C = (b_1, b_2, a_3, b_3)$. Note that C_1, C_2 and C_3 are ordinary Elgamal ciphertexts.

For unclear reasons, the implementation used a prime p with only 256-bits for computations in $\mathbb{Z}_p^*$. You can read the details of the analysis here: `https://arxiv.org/abs/1908.05127`.

(a) Explain how the message M can be recovered from C given the private key (x_1, x_2, x_3)

(b) How long is the private key?

(c) If an attacker can find discrete logarithms in $\mathbb{Z}_p^*$ in time T, how long will take the same attacker to find the message M?

QUESTION 9

Suppose an attacker can break the hash function h used to form a digital signature (RSA or DSA) by finding collisions. How can this lead to attacks on the signature? Is this attack existential or selective?

QUESTION 10

Suppose that RSA signatures are used with a hash function that is not one-way (that is the attacker can invert the hash function). Show how an existential forgery is possible against such a signature: an attacker can form valid signatures from e and n alone.

QUESTION 11

(a) Show that the verification equation for Elgamal signatures works. That is, if (r, s) is a valid Elgamal signature, then $g^m \equiv y^r r^s \bmod p$.

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(b) Similarly check that the verification equation works for DSA signatures.

(c) Similarly check that the verification equation works for ECDSA signatures.

QUESTION 12

Suppose the parameters $p = 23$, $q = 11$, $g = 3$ are used for the DSA signature.

(a) Show that g has order q as required.
(b) If the private key is $x = 5$, what is the public key y?
(c) Compute a valid signature for a message m whose hash value is assumed to be $SHA(m) = 8$.
(d) Show that the verification equation works for your signature.

QUESTION 13

Compare the efficiency of DSA signatures, Elgamal signatures, and RSA signatures assuming that the modulus size is 2048 bits in each case, and that the prime q in DSA signatures is of length 256 bits. Compare:

- the cost of key generation;
- the computation required for signature generation;
- the computation required for signature verification;
- the size of the public keys and signatures.

QUESTION 14

Suppose that the same value of the random k is used to generate two different DSA signatures. Show that this is sufficient for an attacker to find the private signing key. (This was the attack used to break the software verification on the Sony Playstation 3 in 2010 because their implementation used a fixed k. Sony used the elliptic curve version: `https://arstechnica.com/gaming/2010/12/ps3-hacked-through-poor-implementation-of-cryptography/`.)

QUESTION 15

Recall that ECDSA signatures consist of a pair of value (r, s)

(a) Show that if (r, s) is valid ECDSA signature for a message m then $(r, -s)$ is also a valid signature for m.

(b) Show that the ECDSA signature verification equation will be satisfied by the values $(r, s) = (0, 0)$ for *any* message m, as long as other checks are omitted. You can read about why this property led to a huge vulnerability in Java cryptography known as *psychic signatures*.